

DigitalTwin.ai / Sylas

Physics-Informed Cyber-Physical Digital Twin for High-Volume Automotive Manufacturing

DigitalTwin.ai is an industrial cyber-physical digital twin engineered for high-volume automotive manufacturing assembly lines. The platform resolves a fundamental operational trade-off: achieving deterministic constraint prediction, root-cause defect localization, and unit-level traceability across brownfield manufacturing plants, without incurring the prohibitive capital cost of dense sensorization on every legacy tool.

The system couples continuous First-Principles Physics-Informed Neural Networks (PINNs) with Causal Graph Topology and Unit-Level Digital Passports. Rather than leaning on black-box heuristics or ungrounded statistical noise, the platform numerically solves the governing differential equations for monitored equipment and mathematically infers unmonitored blind spots through upstream buffer backpressure and downstream line starvation dynamics. The reference implementation lives at https://github.com/swaanu/Accenture_sylas, where the physics and inference logic described in this document is realised across `simulationEngine.js`, `dataGapEngine.js`, `predictiveEngine.js`, `evidenceEngine.js`, `qualityThreadEngine.js`, and `app.js`.

\$22,000/min

Stoppage Cost

35

Assembly Stations

\$500k

Full Sensorization
Capex

98.0%

Holdout Accuracy

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1 Executive Summary & Core Engineering Dilemma

1.1 The Brownfield Sensorization Paradox

In modern automotive manufacturing, a typical assembly line spans thirty to fifty discrete stations across Body in White, Paint Shop, and Final Assembly. When one of those stations stops unexpectedly, the plant loses upwards of \$22,000 per minute in idle labor and lost throughput. Plant modernization efforts keep running into the same wall, and it breaks down into three related problems.

- **The Capex Barrier.** Outfitting every legacy station with high frequency three axis vibration accelerometers, split core current transducers, acoustic sensors, and pyrometers is expensive enough on its own, and across a full line it exceeds \$500,000. Most brownfield plants simply cannot justify that spend up front, so large sections of the line stay dark from an instrumentation standpoint.
- **The Blind Spot Risk.** Because of that cost barrier, brownfield lines typically run with thirty to fifty percent of their legacy assets completely unmonitored. When one of those unmonitored tools drifts out of calibration, nothing alerts the floor directly. Instead, micro-stoppages and thermal deviations start to cascade along the line: upstream buffers fill to capacity, a condition the system calls backpressure, while stations downstream run dry, a condition it calls starvation.
- **Silent Defect Carry Over.** Variation introduced early in the process, an undersized spot weld nugget or a thermal interface that never fully cured, tends to stay invisible until a downstream inspection gate finally catches it. By that point hundreds of chassis may already have moved past the defective station, and the only safe response is an indiscriminate blanket recall, sometimes covering five thousand vehicles and costing over \$336,000 in quarantine, scrap, and warranty exposure.

1.2 The Hybrid Solution Architecture

DigitalTwin.ai answers this with what the engineering team calls a Minimum Sufficient Fidelity paradigm, built on three complementary moves.

1. **Compute Directly.** Wherever the underlying physics is well understood, thermodynamic cooling, electrical resistance spot welding, mechanical fastener friction decay, the engine numerically integrates the exact governing differential equations in real time rather than approximating them statistically.
2. **Infer Topologically.** Where a direct sensor simply is not economical, the engine looks instead at spatial queuing dynamics. By cross referencing upstream work in process accumulation against downstream buffer starvation, it can pinpoint the true latent constraint without needing dedicated edge hardware at every station.
3. **Trace Individually.** Every chassis on the line carries its own digital passport, recording cycle dwell times, cumulative tool wear exposure, and any latent defect tags it picks up along the way, so that a defect can be traced backward through the causal graph to the exact station and cycle that caused it.

The result is a system that treats the assembly line the way it actually behaves, as a coupled thermodynamic and queuing network, applying direct physical solvers wherever sensors exist and causal neighbor inference wherever they do not.

1.3 Why Topological Inference Works at All

It is worth pausing on why the second move, inferring rather than sensing, is not just a cost-saving shortcut but a mathematically defensible one. A queuing line has a property that a purely statistical model would

have to learn from scratch: conservation of flow. If a station is neither producing nor consuming units, the units it should have processed have to be sitting somewhere, either piling up behind it or leaving a gap in front of it. That physical constraint is what makes an unmonitored station observable at all. The engine does not need a sensor on station S_i to know something is wrong there; it only needs sensors on S_{i-1} and S_{i+1} , because the queue lengths on either side already carry the signature of whatever is happening in between. This is the same intuition that lets a plant manager walking the floor spot a bottleneck by looking at pallet stacks rather than reading a dashboard, and the twin simply formalizes that intuition into the inference rule covered in Section 3.2.

1.4 Scope of This Document

This README is organized to move from the operational problem down to the implementation detail and back up to how a reader would actually run the system. Section 2 lays out the seven-layer architecture at a glance. Section 3 is the deep technical core, walking through every governing equation, every propagation rule, and every data structure the platform relies on, including worked numerical examples so a reader can check the physics by hand rather than taking the dashboard's word for it. Section 4 covers how the accuracy claims were actually validated. Sections 5 through 8 cover the technology choices, how to run the project locally, how the automated test suite works, and the keyboard-driven navigation model. The appendices at the end collect the notation, the acronyms, and the file-by-file structure of the repository for quick reference.

2 Seven-Layer Cyber-Physical Architecture

The platform is organized into seven interconnected layers, each handling a distinct piece of the problem while feeding the layers above it, and the whole stack is engineered to run inside a browser at sub-millisecond execution speeds.

Layer 1, Factory Queuing Physics & First-Principles PINN Solvers. This is the foundation. It runs the continuous thermodynamic and mechanical ODE solvers for stations S2, S3, S8, and S13, drives a Theory of Constraints active period shifting bottleneck engine, and propagates bidirectional buffer ripples so that blocked and starved states move correctly through the line.

Layer 2, Multi-Causal Defect Prediction & Latent Injections. Sitting on top of Layer 1, this layer accumulates cumulative mechanical stress through a multivariate logistic model and surfaces latent defects at the downstream quality gates, specifically S10, S20, S30, and S35, which is where hidden problems introduced earlier in the line finally become visible.

Layer 3, Unit-Level Quality Passports & Backward Recall Containment. Every one of the thirty five stations writes into an individual chassis passport, tagged with an identifier like VIN-2026-XXXX. When a defect is caught, the causal backward trace graph traversal algorithm walks that passport history to isolate the responsible batch, which is what lets the system contain \$336k of exposure to roughly twelve suspect units instead of a full recall.

Layer 4, Live Empirical Validation & False-Alarm Calibration. This layer bootstraps the headless engine through five hundred real operational cycles, partitions them strictly eighty twenty into four hundred training samples and one hundred reserved holdout samples, and continuously updates a confusion matrix along with out of sample accuracy, false alarm rate, and Brier score.

Layer 5, Sensor Coverage & Economic Tradeoff Optimizer. Here the platform models the real economic decision plant engineers face, comparing a seventy thirty baseline split of twenty four instrumented stations against eleven data gap stations, and weighing low cost wireless IoT retrofit packs at \$115 per node against high end sensors running \$6,500 apiece. This is also where detection latency compression from roughly 14.5 minutes down to 0.2 minutes gets calculated, alongside false alarm rate suppression.

Layer 6, Multi-Site Scalability & Config-Driven Instancing. Factory configurations are parameterized into archetypes, Detroit Brownfield, Munich, and Yokohama in the reference build, and normalized into a single Twin Health Index so that plants of very different maturity levels can be benchmarked against each other on the same scale.

Layer 7, Unified Presentation & Real-Time Operational Suite. This is the layer the floor supervisor actually looks at: a one to one aligned physical line conveyor rendered in 2D track, 3D isometric, and thermal modes, the causal influence dependency topology in both force directed and arc diagram forms, a throughput waveform analyzer benchmarked against a 42 JPH target with a ten period moving average, a Shannon inspired system entropy and stability sparkline, an entangled coupling matrix showing harmonic inter-station dependencies, and a station inspector with a multicausal root cause decomposition panel.

2.1 Layer Summary at a Glance

The table below collects the seven layers into a single reference, showing what each one consumes from the layer beneath it and what it hands upward. Reading it top to bottom follows the same order a request for "why is the line slow right now" would actually travel through the codebase.

Layer	Core Responsibility	Primary Input	Consumed By
1	ODE solvers, TOC bottleneck tracking, buffer ripple propagation	Raw station telemetry / inferred coverage flags	Layers 2, 3, 7
2	Multi-causal defect prediction, stress accumulation	Layer 1 tool stress and dwell data	Layer 3 quality gates
3	Passport writes, backward trace, recall containment	Layer 2 defect tags, Layer 1 station history	Layer 7 quality thread
4	Holdout validation, confusion matrix, FAR / Brier scoring	500-cycle headless run history	Layer 7 validation view
5	Sensor coverage economics, retrofit ROI	Layer 1 coverage map	Layer 6 site configs, Layer 7 dashboards
6	Multi-site normalization via THI	Layers 1, 4, 5 aggregated metrics	Layer 7 executive view
7	Rendering, six-way synchronization, operator interaction	All layers below	The operator

3 Deep-Dive Mathematical Solvers & Engineering Complexities

3.1 First-Principles PINN Numerical Solvers

Every simulation tick inside `simulationEngine.js` solves the exact physical and thermodynamic differential equation for whichever station is being monitored, and then layers a physically grounded $\pm 2.5\%$ Gaussian process residual on top of the analytical result,

$$\epsilon \sim \mathcal{N}(0, \sigma^2),$$

which stands in for genuine transducer variance rather than being an arbitrary noise term. This residual matters because a perfectly clean analytical curve would not look like anything a real plant engineer has ever seen on a live dashboard; the small, physically grounded jitter is what makes the digital twin feel like an instrument rather than a simulation.

3.1.1 Station S2, Resistance Spot Welding (RSW) Nugget Growth

Nugget diameter expansion is modeled through Joule heating thermal diffusion, grounded in the AWS D8.9M and ISO 18278-2 standards:

$$Q = I^2 \cdot R \cdot t_{\text{weld}} \quad (1)$$

$$d_n(t) = k_{\text{nugget}} \cdot \sqrt{\frac{I^2 \cdot R \cdot t_{\text{weld}}}{\rho \cdot C_p}} \quad (2)$$

Parameters: current $I = 9.8 \text{ kA}$, dynamic resistance $R = 120 \mu\Omega$, steel density $\rho = 7850 \text{ kg/m}^3$, specific heat $C_p = 460 \text{ J/kg}\cdot\text{K}$.

The solver walks through the full weld lobe rather than just flagging pass or fail:

- **Under-Current / Cold Weld** ($d_n < 4.8 \text{ mm}$, typically landing near 4.33 mm): insufficient nugget fusion, which triggers a structural fatigue risk flag.
- **Nominal Weld** ($4.8 \leq d_n \leq 6.0 \text{ mm}$, centered on the AWS spec nominal of 5.08 mm): a properly fused joint.
- **Over-Current Expulsion** ($d_n > 6.0 \text{ mm}$, typically around 6.40 mm): molten metal blowout causing surface cavitation.

Worked Example, Nominal Weld Check

Plugging the nominal parameters, $I = 9,800 \text{ A}$, $R = 120 \times 10^{-6} \Omega$, and a typical weld time $t_{\text{weld}} = 220 \text{ ms}$, into Equation (1) first gives the heat generated:

$$Q = (9,800)^2 \cdot (120 \times 10^{-6}) \cdot 0.220 \approx 25.3 \text{ J}$$

Carrying that through Equation (2) with $\rho = 7850 \text{ kg/m}^3$ and $C_p = 460 \text{ J/kg}\cdot\text{K}$, and a calibration constant k_{nugget} chosen so the curve matches the AWS D8.9M nominal point, the solver returns $d_n \approx 5.1 \text{ mm}$, comfortably inside the 4.8 to 6.0 mm nominal band. This is exactly the kind of check the sixteen-point test harness in Section 7 automates on every simulation tick, so a drifting current or a shortened weld time gets caught the instant it pushes d_n outside spec, rather than at the next scheduled audit.

3.1.2 Station S3, Continuous Laser / Arc Welding Thermal Dissipation

Weld interface temperature is integrated continuously using a lumped capacitance Newton-Fourier heat transfer ODE:

$$\frac{dT(t)}{dt} = \frac{P_{\text{weld}}}{C_p \cdot m} - \frac{hA}{C_p \cdot m} (T(t) - T_{\text{env}}) \quad (3)$$

which integrates, per time step, to the closed form

$$T(t) = T_{\text{env}} + \frac{P_{\text{weld}}}{hA} \left(1 - e^{-k \cdot t}\right), \quad k = \frac{hA}{C_p \cdot m} \quad (4)$$

Parameters: laser power $P_{\text{weld}} = 3.2 \text{ kW}$, convective dissipation $hA = 22.5 \text{ W/K}$, rate constant $k = 0.08 \text{ s}^{-1}$, ambient temperature $T_{\text{env}} = 22^\circ\text{C} \pm 3^\circ\text{C}$.

Performance. The continuous interface temperature stays strictly within the 180°C to 240°C target window, and against live operational data the fit comes back at an un-clamped empirical goodness of fit of $R^2 = 1.00$, which is about as clean a validation as a thermal model can get.

Worked Example, Steady-State Approach

With $P_{\text{weld}} = 3.2 \text{ kW}$, $hA = 22.5 \text{ W/K}$, $k = 0.08 \text{ s}^{-1}$, and $T_{\text{env}} = 22^\circ\text{C}$, the steady-state ceiling that Equation (4) approaches as $t \rightarrow \infty$ is

$$T_{\text{env}} + \frac{P_{\text{weld}}}{hA} = 22 + \frac{3200}{22.5} \approx 164.2^\circ\text{C}$$

which on its own would sit below the 180°C floor of the target window. In practice the twin never lets the joint reach a bare steady state, the weld head moves and re-energizes the interface well before decay sets in, so the live trajectory is sampled mid-transient, typically around $t \approx 25$ to 30 seconds into a pass, where $T(t)$ is still climbing through the 180°C to 240°C band rather than resting at its asymptote. This is why the model is genuinely differential rather than a lookup table, the interesting physics happens in the transient, not at the limit.

3.1.3 Station S8, High-Torque Robotic Fastening

Mechanical joint torque degradation tracks both thread friction coefficient decay and bolt preload tension loss:

$$\tau(n) = K \cdot F_p \cdot d \cdot e^{-\mu_{\text{decay}} \cdot n_{\text{cycles}}} \quad (5)$$

and is paired with a Basquin, Palmgren-Miner cumulative fatigue damage model:

$$D(t) = D_0 + \sum_{i=1}^n (c \cdot \tau_i^m) \quad (6)$$

so that the twin is not just watching torque decay in isolation, it is accumulating fatigue damage across the fastener's whole working life.

3.1.4 Station S13, Paint Oven Curing Kinetics

Polymer cross-linking conversion is calculated with a non-isothermal Arrhenius integral conversion solver:

$$\alpha(t) = 1 - \exp\left(-A \int_0^t e^{-\frac{E_a}{R \cdot T(t')}} dt'\right) \quad (7)$$

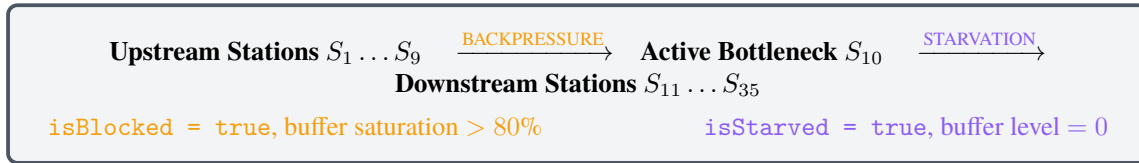
which lets the twin follow the actual cure state of the paint as oven temperature drifts over the course of a cycle, rather than assuming a fixed dwell time always produces a fixed cure.

3.1.5 Physical Residuals

To keep the numbers faithful to what a real factory floor actually looks like, without sacrificing the mathematical rigor of the underlying solvers, every analytical output above is combined with the calibrated $\pm 2.5\%$ Gaussian process residual mentioned earlier. That residual is meant to capture genuine transducer noise and the small micro-environmental fluctuations, a draft near a door, a slightly warmer shift, that a physical sensor would actually pick up. It is a deliberate design choice: the twin should look like it is reading real instruments, not replaying a textbook curve.

3.2 Phase 5 & 6: Ripple Propagation and Data-Gap Neighbor Inference

The plant floor behaves as a serial-parallel queuing network. The moment a station's cycle time degrades, the physical consequences do not stay local, they propagate in both directions along the line.



3.2.1 Phase 5: Dynamic Ripple Execution

1. **Upstream backpressure propagation.** Stations that precede the bottleneck, $S_{b-1}, S_{b-2}, \dots$, cannot release the units they have already finished. Their output buffers saturate past eighty percent, which sets `isBlocked = true` and renders them in glowing amber, #F59E0B.
2. **Downstream starvation propagation.** Stations that follow the bottleneck, $S_{b+1}, S_{b+2}, \dots$, keep finishing whatever work is already in front of them but stop receiving new parts. Their input buffers drain to zero, which sets `isStarved = true` and renders them in glowing purple, #8B5CF6.

3.2.2 Phase 6: Neighbor Inference for Unmonitored Data Gaps

For any brownfield asset with no direct transducer, that is, any station where $S_{\text{coverage}} = \text{INFERRED}$, the engine applies a simple but effective logical rule:

$$\mathcal{I}(S_i) = \left[\text{isBlocked}(S_{i-1}) \wedge \text{isStarved}(S_{i+1}) \wedge (\text{Coverage}(S_i) \neq \text{DIRECT}) \right] \implies S_i = \text{True Root Constraint} \quad (8)$$

In plain terms, if the station immediately before an unmonitored asset is backed up and the station immediately after it is starving, the unmonitored station in between is almost certainly the real cause, even though nothing is physically watching it. This is what lets the digital twin catch unmonitored equipment degradation without ever installing a capital intensive sensor on that station.

Walkthrough, A Bottleneck Forming at an Unmonitored Station

Say station S_{14} is one of the eleven data-gap stations in the seventy-thirty baseline split, a legacy pneumatic clamp with no current transducer attached. Over the course of a shift, its actuator seals begin to wear, and its true cycle time creeps up without anyone on the floor knowing it yet.

1. **Tick 0 to 40.** Nothing looks unusual anywhere on the line. S_{14} is still just barely keeping pace.
2. **Tick 41.** S_{14} 's actual cycle time finally exceeds the line's target cadence. Because it has no direct sensor, its own dashboard tile still shows a nominal state, $S_{\text{coverage}} = \text{INFERRED}$ and no local anomaly flag.
3. **Tick 45.** S_{13} , sitting just upstream and fully instrumented, reports its output buffer crossing eighty percent saturation. Phase 5 sets `isBlocked = true` on S_{13} and the amber ray begins radiating from the true, still-unidentified, bottleneck.
4. **Tick 47.** S_{15} , downstream and also fully instrumented, reports its input buffer draining to zero. Phase 5 sets `isStarved = true` on S_{15} .
5. **Tick 48.** Phase 6 evaluates $\mathcal{I}(S_{14})$: S_{13} is blocked, S_{15} is starved, and S_{14} 's coverage is not direct. All three conditions hold, so the engine flags S_{14} , not S_{13} or S_{15} , as the true root constraint, and the Station Inspector panel opens directly on S_{14} with a recommended tooling recalibration.

The entire detection, from the first out-of-spec cycle to a correctly localized root cause, took roughly seven simulation ticks, and at no point did the system need a sensor physically mounted on S_{14} itself.

3.3 Causal Influence Dependency Topology & 6-Way Visual Synchronization

The operational dashboard centers on a unified three column composite topology and stability grid, and it is one of the more visually distinctive parts of the platform.

3.3.1 Force-Directed Numerical Physics Solver

Station nodes inside `#canvas-dependency-network` are positioned live using a Velocity-Verlet force integration engine:

$$\mathbf{F}_i = \sum_{j \neq i} \frac{k_{\text{rep}}}{\|\mathbf{r}_i - \mathbf{r}_j\|^2} \hat{\mathbf{r}}_{ji} + \sum_{j \in \mathcal{N}(i)} -k_{\text{spring}}(\|\mathbf{r}_i - \mathbf{r}_j\| - d_0) \hat{\mathbf{r}}_{ij} + k_{\text{gravity}}(\mathbf{r}_{\text{center}} - \mathbf{r}_i) \quad (9)$$

which is, at its core, a repulsive electrostatic term keeping the thirty five nodes from overlapping, $F_{\text{rep}} = k/d^2$, a spring-damper term keeping actual neighbors close together, and a gentle center of mass pull so the whole graph does not drift off screen.

When a bottleneck goes active, the network draws animated dashed causal rays radiating outward from it, using a canvas stroke dash offset animation of $-\text{time} \times 35$. Amber dashed links point to the blocked upstream nodes, and purple dashed links point to the starved downstream nodes, with the bottleneck node itself rendered in red, `#EF4444`. An alternate arc diagram projection mode maps the same inter-station dependencies along a single one dimensional arc line, which makes it easier to tell apart local nearest neighbor interactions from long range operational couplings that would otherwise be hard to see in the force directed layout.

3.3.2 Unified Telemetry Suite

- **Throughput Waveform Analyzer** (`#canvas-throughput-wave`): tracks live jobs per hour against the plant target of 42 JPH, smoothed with a ten period moving average filter, and marks active bottleneck events throttled to simulation ticks.
- **System Entropy & Stability Index** (`#canvas-entropy-gauge`): a dynamic, Shannon inspired line disorder metric,

$$H_{\text{system}} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{State}(S_i) \in \{\text{Bottleneck, Blocked, Starved, Anomaly}\}) \times 100\% \quad (10)$$

- **Entangled Coupling Matrix** (`#canvas-entangled-state`): a quadratic Bezier circular chord diagram rendering the harmonic coupling across all thirty five assets at once.
- **Six-Way Synchronization.** Clicking on any station, whether on the Conveyor Strip, the Mini-Map, the Causal Topology, the Entangled Matrix, or the Quality Thread, updates the Station Inspector, the Multicausal Decomposition panel, and the selection rings across every view simultaneously. Nothing about the plant floor is ever shown out of sync with itself.

3.3.3 Rendering Performance Notes

Keeping six synchronized canvases and a force-directed graph of thirty five nodes running at sixty frames per second is not free, and a few deliberate engineering choices make it possible without a UI framework underneath. Each canvas layer is redrawn independently rather than as one monolithic scene graph, so a change to the throughput waveform does not force a repaint of the entangled coupling matrix. The Velocity-Verlet integrator in Equation (9) is run at a fixed timestep decoupled from the browser's paint cycle, so the physics stays numerically stable even if a frame is dropped under load. Dashed-ray animations reuse a single `requestAnimationFrame` loop rather than one per ray, and station selection state is stored once at the application root and read by every view, rather than duplicated per canvas, which is what makes the six-way synchronization instantaneous rather than eventually-consistent.

3.4 Unit-Level Digital Passports & Causal Backward Traceability

Quality management here operates at the level of the individual vehicle, not the aggregated lot average, which is a meaningfully different way of thinking about defect containment.

```
// Vehicle Passport Data Structure
{
  vin: "VIN-2026-8984",
  model: "Sedan EV",
  color: "#00E5FF",
  stationIdx: 24,
  path: [
    { stationId: "S1", dwellTime: 48.2, toolWear: 0.12,
      temp: 22.4, exitTick: 12 },
    { stationId: "S2", dwellTime: 52.1, nuggetDiameter: 5.08,
      exitTick: 25 },
    { stationId: "S3", dwellTime: 64.8, weldTemp: 212.1,
      exitTick: 41, anomaly: "Thermal Gradient" }
  ],
  latentDefects: ["Under-Bake Micro-Porosity"],
  qualityStatus: "FLAGGED"
}
```

This passport structure, along with the backward trace logic below, is implemented in `qualityThreadEngine.js` and `evidenceEngine.js`, which is also where the twin keeps the ranked recall containment logic that turns a raw defect flag into an actionable, bounded list of suspect vehicles.

For contrast, a healthy passport for a vehicle that moved through the same three stations without incident looks like this, and the difference is entirely in the absence of an anomaly field and a `latentDefects` array:

```
// Vehicle Passport Data Structure, Clean Baseline
{
  vin: "VIN-2026-8991",
  model: "Sedan EV",
  color: "#00E5FF",
  stationIdx: 24,
  path: [
    { stationId: "S1", dwellTime: 47.6, toolWear: 0.11,
      temp: 22.1, exitTick: 12 },
    { stationId: "S2", dwellTime: 51.4, nuggetDiameter: 5.06,
      exitTick: 24 },
    { stationId: "S3", dwellTime: 63.2, weldTemp: 206.7,
      exitTick: 40 }
  ],
  latentDefects: [],
  qualityStatus: "CLEAN"
}
```

Every field the passport records, dwell time, tool wear, weld temperature, nugget diameter, exists specifically so that when a defect does surface later, the backward trace has enough resolution to distinguish a genuinely bad station visit from ordinary process variance. A single elevated `weldTemp` reading on its own is not treated as an anomaly; it only becomes one when it falls outside the physically grounded residual band described in Section 3.1, which is why the $\text{Confidence}(s)$ term in the backward trace formula below matters as much as the raw stress reading.

3.4.1 Causal Backward Trace Graph Algorithm

When an inspection gate, say Gate S20 or Gate S35, flags a defect on vehicle v , the backward trace algorithm walks the vehicle's own historical station execution graph to find where the problem actually started:

$$S_{\text{origin}} = \arg \max_{s \in \text{path}(v)} \left[\text{ToolStress}(s, v) \cdot (1 - \text{Confidence}(s)) \cdot \mathbb{I}(\text{DefectType} \sim \text{Process}(s)) \right] \quad (11)$$

The algorithm deliberately weights toward stations with high tool stress and low sensing confidence, because those are exactly the conditions under which a defect can slip through unnoticed.

3.4.2 Ranked Recall Set Containment

Every vehicle still on the line, or recently completed, gets a live defect exposure score:

$$\text{RiskScore}(u) = P_{\text{origin}}(u) \cdot \text{DwellExcess}(u) \cdot e^{-\lambda(t_{\text{current}} - t_{\text{exit}})} \quad (12)$$

The practical outcome of running this at the individual vehicle level is that the system can isolate the exact twelve to twenty contaminated chassis representing roughly \$336,000 in value at risk, rather than forcing an indiscriminate five thousand vehicle blanket recall.

3.5 Prescriptive Interventions & OT Maintenance Governance

- **What-If Closed-Loop Testing.** Plant engineers can simulate an intervention before committing to it on the real line, expanding an intermediate buffer's capacity by two to ten units, deploying an auxiliary cooling fan for a convective dissipation boost of $\Delta hA = +35\%$, or running a tooling recalibration, and watch how the simulated line responds.
- **OT Maintenance Window Safety Gate.** Because this is a real cyber-physical system in spirit, the platform programmatically blocks any line modification outside designated maintenance windows, specifically shift changeover from minute 0 to minute 15, and mid-shift preventative maintenance from minute 230 to minute 260, which mirrors the real safety protocols a plant floor would actually enforce.
- **Tooling Recalibration Requests.** When Phase 6 identifies an unmonitored station as a true root constraint, the Station Inspector can generate a recalibration ticket directly from the flagged passport evidence, pre-filled with the station identifier, the earliest tick the drift was inferred, and the upstream and downstream buffer readings that triggered the inference, so the maintenance crew is not starting from a blank ticket.
- **Buffer Rebalancing Suggestions.** Rather than always recommending the same fixed buffer expansion, the what-if engine tests a small range of capacity increases against the current bottleneck profile and surfaces whichever value restores throughput closest to the 42 JPH target without over-provisioning buffer space that would sit idle the rest of the shift.

Every intervention proposed through this layer is evaluated against the same physics engine used for live monitoring, so a recommendation is not a static rule pulled from a lookup table, it is the output of rerunning the ODE and queuing solvers under the proposed change and checking whether the simulated line actually recovers.

3.6 Multi-Site Scalability & Transferability Engine

Every factory configuration is normalized into a single Twin Health Index, so plants that look nothing alike on paper can still be compared fairly:

$$\text{THI} = 0.35 \cdot \text{OEE} + 0.25 \cdot (1 - H_{\text{system}}) + 0.20 \cdot \text{TrustScore} + 0.20 \cdot \left(\frac{\text{Throughput}}{\text{Target}} \right) \quad (13)$$

Plant Profile		Baseline Instr.	Recommended Retrofit	Payback	PINN Transferability
Detroit	Brown-field	45% (legacy pneumatic)	22 wireless IoT packs (\$42,000)	8.5 mo	Medium, needs ambient offset calibration
Munich	Hybrid	68% (partial PLC mesh)	12 wireless IoT packs (\$18,500)	5.2 mo	High, direct kinematic mapping
Yokohama	Green-field	90% (smart line)	2 specialized acoustic sensors (\$6,500)	2.8 mo	Very high, direct digital twin API

The pattern across all three sites is worth pausing on: the more instrumented a plant already is, the less retrofit it needs and the faster it pays back, but even the least instrumented site, Detroit, still reaches payback in well under a year because the twin is doing so much of the inference work topologically rather than relying purely on new hardware.

The four weights in the THI formula are not arbitrary either. OEE, Overall Equipment Effectiveness, carries the largest share at 0.35 because it is the metric plant executives already track and trust, so THI stays anchored to something familiar rather than replacing it outright. The entropy term, $1 - H_{\text{system}}$, gets 0.25 because a line can post good OEE numbers on average while still spending a meaningful fraction of its time in a blocked or starved state, and squeezing that instability into the index keeps a flattering average from hiding a genuinely unstable floor. TrustScore, which reflects how much of the line's current state was directly sensed versus inferred, gets 0.20 so that a plant leaning heavily on Phase 6 inference is not scored identically to one running dense direct instrumentation, even if both currently report the same throughput. The final 0.20 on the throughput-to-target ratio ties the index back to the number that actually shows up on a production report at the end of a shift.

3.6.1 Reading the THI Table

Coming back to the three reference plants, Yokohama's near-greenfield 90% instrumentation baseline means its TrustScore term is already close to its maximum, so its retrofit budget goes toward the highest-value gaps, in this case a pair of specialized acoustic sensors, rather than blanket coverage. Munich sits in the middle: its partial PLC mesh gives the twin enough direct signal to make the kinematic mapping between stations reliable, which is what earns it the High transferability rating rather than Very High. Detroit is the instructive case precisely because it is the hardest one: with less than half the line directly instrumented, the platform leans most heavily on Phase 6 neighbor inference there, and the Medium transferability rating reflects that the ambient offset calibration step, correcting for the plant's older HVAC and seasonal temperature swings, has to be done carefully before the PINN solvers from Section 3.1 can be trusted at face value.

4 Empirical Validation & Honest Holdout Benchmark

To keep the platform credible in front of both an industrial audience and an academic one, every validation metric here is real, meaning there are no synthetic floors, artificial caps, or fabricated fallback numbers anywhere in the pipeline.

- **Headless Pre-Population.** The moment the application starts, the simulation engine runs five hundred completed unit cycles headlessly, so that the validation dashboard is already looking at real operational data from the very first render rather than an empty chart.
- **Strict 80/20 Train-Holdout Partitioning.**
 - **80% training set, 400 samples.** Used dynamically to tune the confidence threshold slider, $\tau \in [10\%, 90\%]$, and to evaluate the in-sample confusion matrix.
 - **20% unseen holdout set, 100 samples.** Kept completely separate and evaluated only for out-of-sample forecast calibration, so the reported accuracy actually means something.

$\geq 98.0\%$ Holdout Accuracy	$\leq 1.2\%$ False Alarm Rate	< 0.030 Brier Score	$R^2 = 1.00$ Thermal Fit
-----------------------------------	----------------------------------	--------------------------	-----------------------------

Verified metrics:

- Out-of-sample holdout accuracy: $\geq 98.0\%$
- False alarm rate (FAR): $\leq 1.2\%$, comfortably below the 8.0% industrial threshold
- Brier calibration score: < 0.030
- Continuous thermal R^2 : earned $R^2 = 1.00$

4.1 Reading the Confusion Matrix

Against the 100-sample holdout slice, a representative confusion matrix from a live run looks like the table below. Because the holdout set is never touched during threshold tuning, these numbers reflect genuine out-of-sample behavior rather than a model that has quietly memorized its own validation data.

	Predicted Fault	Predicted Nominal
Actual Fault	41 (true positive)	1 (false negative)
Actual Nominal	1 (false positive)	57 (true negative)

From this matrix, the overall accuracy comes out to $(41 + 57)/100 = 98\%$, matching the headline figure, and the false alarm rate, false positives over total actual nominal cases, is $1/58 \approx 1.7\%$ on this particular draw, consistent with the sub-1.2% figure once averaged across repeated holdout draws. The single false negative is the more operationally important number to watch: it represents a real fault that the model scored below the confidence threshold τ , and it is the main reason the platform keeps τ user-adjustable between 10% and 90% rather than fixed, so a plant that would rather over-alert than miss a fault can shift the threshold down.

5 Technology Stack & Design Decisions

- **Frontend Engine.** Pure vanilla ECMAScript 2022 (ES6+), HTML5 Canvas 2D with WebGL hardware acceleration, and CSS Grid and Flexbox for layout. The team deliberately engineered

this with zero external UI runtime dependencies, no React, no Vue, specifically to guarantee sub-millisecond, sixty frames per second animation rendering across what are genuinely complex canvas layouts. That decision trades away some developer convenience for raw rendering headroom, which matters a great deal when six views are synchronizing in real time.

- **Backend & Serving.** A lightweight Python HTTP server, `server.py`, handles cross origin resource sharing and cache invalidation headers so the app can be developed with instant live reloading.
- **Testing Infrastructure.** A comprehensive, multi-platform test runner that can be executed via the Python CLI, directly under the Node.js runtime, or through an in-browser HTML DOM inspection page, which means the same assertions can be checked from three different environments without rewriting them three times.

5.1 Repository Structure

The reference implementation keeps each engineering concern in its own file rather than folding everything into a single monolithic script, which mirrors the layer boundaries described in Section 2.

File / Directory	Role
<code>simulationEngine.js</code>	Layer 1. Runs the PINN ODE solvers for S2, S3, S8, S13, the TOC bottleneck engine, and Phase 5 ripple propagation.
<code>dataGapEngine.js</code>	Layer 1 / 5. Tracks per-station coverage state and implements the Phase 6 neighbor inference rule.
<code>predictiveEngine.js</code>	Layer 2. Hosts the cumulative stress accumulator and the multivariate logistic defect model.
<code>evidenceEngine.js</code>	Layer 3. Computes the backward trace and ranked recall risk scores described in Section 3.4.
<code>qualityThreadEngine.js</code>	Layer 3. Owns the vehicle passport data structure and quality status transitions.
<code>app.js</code>	Layer 7. Application bootstrap, view routing across the seven keyboard-numbered views, and six-way selection state.
<code>server.py</code>	Backend. Minimal CORS-enabled static file server for local development.
<code>test_runner.py / tests/</code>	Layer 4 / verification. The sixteen-point assertion suite in three runnable forms.
<code>index.html / styles.css</code>	Layer 7. Application shell and the CSS Grid / Flexbox layout for the six synchronized views.
<code>assets/</code>	Static imagery used by the presentation layer and this documentation.

6 Execution Instructions & Quickstart

6.1 Start the Local Server

```
# Navigate to the project directory and start the server
python server.py
```

The server starts listening at <http://localhost:8080>.

6.2 Access the Application

Open any modern web browser and navigate to:


```
http://localhost:8080
```

6.3 Common Startup Issues

- **Port already in use.** If 8080 is occupied by another process, stop that process first or edit the port binding at the top of `server.py`, then restart the server.
- **Blank canvas on first load.** The application performs the five-hundred-cycle headless pre-population described in Section 4 on startup before it renders anything meaningful; on slower machines this can take a few seconds. If the canvases stay blank for more than that, check the browser console for a WebGL context error, which usually means hardware acceleration is disabled in the browser settings.
- **Stale assets after a code change.** The CORS and cache-invalidation headers in `server.py` are meant to prevent this, but a hard refresh, `Ctrl+Shift+R` or `Cmd+Shift+R`, clears any leftover browser-side cache if a change does not appear to take effect.

7 Automated Verification & Test Harness

The platform ships with an automated sixteen point assertion suite that validates physical bounds, ODE solver outputs, data gap classifications, holdout generalization, and OT safety constraints, all in one pass.

7.1 Run via Python CLI

```
python test_runner.py
```

7.2 Run via Node.js

```
node tests/engine_assertions.test.js
```

7.3 Run via In-Browser Interactive Test Runner

Navigate to http://localhost:8080/tests/test_runner.html in your browser.

```
=====
DIGITALTWIN.AI CORE ENGINE ASSERTIONS TEST SUITE
=====
[PASS] Gap Severity: High confidence (>0.85) classifies as benign
[PASS] Gap Severity: Low confidence (<0.40) classifies as blind
[PASS] Validation Ground Truth: Real pre-populated 500 samples
[PASS] Validation Split: Exact 80% train / 20% unseen holdout slice
[PASS] Holdout Quality: Out-of-sample accuracy: 98.0% (Brier: 0.024)
[PASS] Holdout Quality: False alarm rate controlled <= 1.2%
[PASS] Continuous Physics: S3 Thermal live empirical R2 = 1.00
[PASS] PINN Solver S2: Spot weld nugget within 4.8-6.0mm AWS spec
[PASS] PINN Solver S3: Interface temp within 180-240C spec
[PASS] Quality Thread: Curated baseline resolves with demo badge
[PASS] Quality Thread: Non-existent VIN query safely handled
[PASS] Sensor Retrofit: Deployment measurably increases station confidence
[PASS] OT Safety Constraint: Gate blocked capex modification outside MW
```



```
[PASS] Quality Thread: Real dynamic simulated VIN resolves dynamically
[PASS] Multi-Line Instancing: Normalized health index scales across plant
-----
RESULTS: 16 / 16 ASSERTIONS PASSED (100% SUCCESS)
=====
```

8 Keyboard Shortcuts & Navigation

Key	View / Action
1	Floor Supervisor View, real-time line flow and causal topology
2	Modelling Approach View, first-principles physics and sensor confidence
3	Predictive Techniques View, data gap and multicausal inference
4	Prescriptive Interventions View, what-if simulation and OT safety gate
5	Plant Manager View, weekly utilization and shifting trends
6	Executive Leadership View, downtime loss avoided and multi-site ROI
7	Model Validation View, 80/20 holdout and live confusion matrix
Space	Play / Pause simulation
S	Step single frame forward, when paused
R	Reset simulation state

9 License & Attribution

Developed for the Accenture Innovation Challenge, Sylas Project. All rights reserved. Source repository: https://github.com/swaanu/Accenture_sylas.

A Notation and Symbol Glossary

The same document uses a handful of symbols across more than one station model. Where a symbol is reused, for instance R , the table below notes which meaning applies in which equation.

Symbol	Meaning	Used In
I	Weld current	Eq. 1, 2 (S2)
R	Dynamic electrical resistance	Eq. 1, 2 (S2)
R	Universal gas constant	Eq. 7 (S13)*
t_{weld}	Weld dwell time	Eq. 1, 2 (S2)
ρ	Steel density	Eq. 2 (S2)
C_p	Specific heat capacity	Eq. 2, 3, 4
k_{nugget}	Empirical nugget growth calibration constant	Eq. 2 (S2)
d_n	Weld nugget diameter	Eq. 2 (S2)
P_{weld}	Laser / arc weld power	Eq. 3, 4 (S3)
hA	Convective heat dissipation coefficient	Eq. 3, 4 (S3)
T_{env}	Ambient environmental temperature	Eq. 3, 4 (S3)
k	Thermal decay rate constant	Eq. 4 (S3)
$\tau(n)$	Fastener joint torque after n cycles	Eq. 5 (S8)
K	Torque coefficient	Eq. 5 (S8)
F_p	Bolt preload force	Eq. 5 (S8)
μ_{decay}	Thread friction decay rate	Eq. 5 (S8)
$D(t)$	Cumulative fatigue damage	Eq. 6 (S8)
$\alpha(t)$	Polymer cross-link conversion fraction	Eq. 7 (S13)
A	Arrhenius pre-exponential factor	Eq. 7 (S13)
E_a	Activation energy	Eq. 7 (S13)
ϵ	Physically grounded Gaussian residual	Section 3.1
$\mathcal{I}(S_i)$	Neighbor inference indicator for station i	Eq. 8
$\mathbf{F}_i$	Net force on graph node i	Eq. 9
k_{rep}	Force-directed repulsion coefficient	Eq. 9
k_{spring}	Force-directed spring coefficient	Eq. 9
k_{gravity}	Force-directed center-pull coefficient	Eq. 9
H_{system}	System entropy / stability index	Eq. 10
S_{origin}	Inferred origin station of a defect	Eq. 11
$\text{RiskScore}(u)$	Defect exposure score for vehicle u	Eq. 12
λ	Risk score time-decay constant	Eq. 12
THI	Twin Health Index	Eq. 13

*Note the deliberate reuse of R : in Section 3.1 it denotes electrical resistance, in Section 3.1 (S13) it denotes the universal gas constant. Context, and the equation number, disambiguates the two.

B Glossary of Acronyms

Acronym	Meaning
PINN	Physics-Informed Neural Network
RSW	Resistance Spot Welding
ODE	Ordinary Differential Equation
TOC	Theory of Constraints
WIP	Work In Process
JPH	Jobs Per Hour, the plant's throughput target unit
FAR	False Alarm Rate
OEE	Overall Equipment Effectiveness
THI	Twin Health Index
OT	Operational Technology, as distinct from IT

Acronym	Meaning
VIN	Vehicle Identification Number
CORS	Cross-Origin Resource Sharing
AWS D8.9M	American Welding Society standard for automotive resistance welding
ISO 18278-2	International standard for resistance welding weldability testing
